

RYAN SUNGMO KWON

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EDUCATION

University of Michigan - Ann Arbor

Ann Arbor, MI

B.S. Computer Science

August 2022 – Expected April 2026

- **GPA:** 3.835/4.0
- **Relevant Coursework:** Computer Vision, Intro to Machine Learning, Data Structures and Algorithms, Intro to Computer Organization, Discrete Math, Linear Algebra, Differential Equations, Multivariable and Vector Calculus

PROFESSIONAL EXPERIENCE

Statistics Online Computational Resource (SOCR)

Ann Arbor, MI

Research & Software Engineering Intern & Project Lead

January 2024 – Current

- Conducted research under Professor Ivo Dinov and deployed the SOCR AI Bot; currently writing for SOCR's AI Brain Data Generator (GAN & Diffusion) for the International Journal of Biomedical Imaging with Achu Shankar as a 2nd author
- Fine-tuned Llama-3 for medical queries and code-generating/distribution plotting purposes utilizing Q-Lora (PEFT) and 14000+ rows of assorted medical text data with multiple GPUs from the University of Michigan's HPC Cluster
- Engineered a generative AI model that leverages the OpenAI, Google Gemini, and DALL·E API, integrated with RShiny, significantly enhancing the application's ability to produce dynamic, accurate clustering and visualization results, thereby optimizing data-driven decision-making processes

Deloitte Consulting - Monitor Deloitte

Seoul, South Korea

Research Assistant Intern

June 2023 – August 2023

- Conducted extensive research for The Korean Chamber of Commerce & Industry on approximately 100 countries, including USA, UK, Brazil, Equatorial Guinea, and Haiti, to assess the possibility of their support in the 2030 World Expo in Busan.
- Created comprehensive country profiles and reports highlighting the economic, technological, cultural, and political factors that Korea could support to influence the countries' decisions to back Busan's bid for the 2030 World Expo.
- Collected and managed data and conducted research on green hydrogen policies, regulations, and market conditions in Australia, United Kingdom, and Chile to assess the potential and enhance the prospect for Korea Hydro & Nuclear Power's overseas expansion of their renewable hydrogen business in those countries.

Gauchocourses

Santa Barbara, CA

Frontend Team

October 2022 – February 2023

- A software application designed to manage course schedules specifically for students attending UCSB and other universities
- Partnered up with another member of the organization and shadowed the development and bug fixes of the frontend of the application using various coding languages and frameworks including CSS, HTML, Javascript, and Vue.js

RESEARCH & PROJECTS

Yonsei University MIR Lab

South Korea

Research Intern (Inverse Ops & AI Safety)

June 2024 – Current

- Trained and utilized VQVAE models to effectively compress images into latent representations for downstream processing
- Collaborated on employing diffusion models to reverse complex random noises, such as Gaussian noise, blur, and resizing
- Applied the paper "Circuit Breakers" and currently experimenting to obtain the safety vector of LLMs through finetuning with automated red teaming datasets and subtracting the original model in the vector space.

Dylan (Google X MHacks Hackathon)

Ann Arbor, MI

Backend (AI) Developer

May 2024

- Code-assistant AI utilizing Google's Gemini 1.5 Pro API and PyGitHub that combines every file and data from a Github repository to comprehend the code's structure and rationale, and produces in-depth contexts to add layers of personalization.
- Utilized ChromaDB to efficiently store and retrieve the embeddings of the code to precisely train and perceive the prompts.

ACTIVITIES & LEADERSHIP

University of Michigan Solar Car Team

Ann Arbor, MI

Strategy Division, Member

September 2023 – June 2024

- Machine Learning project by developing the Ant Colony Optimization Reinforcement Learning algorithm to simulate and produce the most efficient speed incorporated with various environmental and kinematic factors for sectors of the race.
- Developed a car simulator program calculating the estimated race time of the solar car among with various factors including resistive force, wind heading, and bearing.

UCSB Data Science Club: Optimal Business Locations in LA

Santa Barbara, CA

Project Pipeline, Project Team Leader

September 2023 – June 2023

- Developed a machine learning algorithm with the Bayesian Weighted Average formula and KNN algorithm to determine the optimal restaurant location in LA by leveraging numerous factors including crime rates and population density by race.
- Employed the Beautiful Soup package for web scraping HTML attributes to extract pertinent data, and handled over 700,000+ rows of data by utilizing the Google Map Geocoding API for a more efficient and enhanced analysis of those data.

SKILLS

Skills: C/C++, Python (Pytorch), R, Machine Learning (NLP and Computer Vision), Java, SQL

Languages: English (native), Korean (native), Chinese (HSK Level 3)